

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Two-variable data: Models and scatterplots

04

2 answers · Rationale-rich · Teach from doc

Q1**A**

A. $y = 2.8 + 1.7x$

Choice A is correct. The line of best fit shown intersects the y -axis at a positive y -value and has a positive slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y -axis at a y -value of a and has a slope of b . Of the given choices, only choice A represents a line that intersects the y -axis at a positive y -value, 2.8, and has a positive slope, 1.7.

Choice B is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice C is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

Choice D is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value, and has a negative slope, not a positive slope.

Q2**B**

B. 125

Choice B is correct. According to the graph, when the car is traveling at 50 miles per hour, the braking distance is approximately 225 feet, and when the car is traveling at 30 miles per hour, the braking distance is approximately 100 feet. The difference between these braking distances is $225 - 100$, or 125 feet.

Choice A is incorrect and may result from finding the braking distance for 20 miles per hour, the difference between the given speeds. Choice C is incorrect and may result from subtracting the speed from the braking distance at 50 miles per hour. Choice D is incorrect and may result from finding the difference in the braking distances at 60 and 20 miles per hour.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

